

Safe Place VR: A Guided Imagery-Based Virtual Reality Trauma Stabilization Technique

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ABSTRACT

One of the techniques used in trauma stabilization stage is ‘Safe Place’ which aims to establish a sense of safety by guiding patients to create a vivid mental imagery. Existing research has highlighted both the advantages and potential of using VR for trauma intervention, as well as some limitations. In this study, we present the ‘Safe Place VR’. Specifically, we provide three pre-set virtual environments, three different time-of-day lighting settings, and several comfort items representing various senses that may be linked to positive emotions or memories, guiding patients to create their own safe place within the VR environment.

Index terms: Virtual reality, guided imagery, trauma stabilization.

1 INTRODUCTION

Virtual reality (VR) can offer realistic visualization and immersive experience, providing a safe and controllable environment that enhances the possibilities for psychological research and treatment. Recently, therapists have recognized the benefits of VR for guided imagery, which engages both the imagination and the senses.

Here, we focus on the safe place trauma stabilization technique which used guided imagery. We provide three virtual environments (bedroom, beach, and mountain meadow), three different period lighting settings (morning, noon and afternoon), and the option of objects which may linked to positive memories and emotions (such as photo frames, cushion, and candle) to guide users to create their own safe place. Finally, we allow users to name and keep their safe place as picture to return to it any time in the future.

2 RELATED WORK

2.1 Safe Place in Trauma Treatment

Trauma stabilization is the initial phase of trauma treatment with the main goal of establishing a sense of safety and control [1]. One of the stabilization techniques is the *Inner Safe Place* (or *Safe Space*). Therapists generally guide patients to create a mental imagery of a safe place through guided imagery (GI) method [1]. By vividly imagining scenes that bring them a sense of calm or recalling positive memories, patients can prevent painful experiences and intrusive traumatic memories from resurfacing [2]. In addition, patients with trauma symptoms may gain a sense of safety from their comfort items, such as blankets or teddy bears.

These comfort items can help them relieve anxiety and develop the control skills to reduce distress [3].

2.2 Safe Place with Virtual Reality

Using digital audio files on smartphones to self-practice the safe place technique without a therapist has been considered helpful and effective for patients [4]. However, some patients may have difficulty imagining a place in their minds [4]. Several VR stabilization techniques have been developed and reported to be effective in reducing psychological distress associated with experiencing traumatic events [5].

Compared to previous studies that implemented the safe place technique by providing several fully preset VR environment [6], some therapists recommend offering a partially preset virtual environment that allows users to personalize it by adding preset objects and creating their own unique virtual space [7].

Other studies explored the combining of art therapy to help individuals understand their emotions and thoughts for the psychological safety [8]. One study reported the positive effects of art-making experiences of creating their safe space in 3D virtual environment. However, due to the lack of artistic skills required to create an aesthetically art work, some participants did not feel self-satisfied in the VR creative process [8].

3 SAFE PLACE VR OVERVIEW

The contents of Safe Place VR is based on a guided imagery script [9], and it was subsequently adapted and translated into Korean by our collaborating psychological experts, with a total duration of approximately 8 minutes. It was developed using Unity 2021.3.0f1 and tested with an HTC Vive Eye head-mounted device (HMD).

3.1 Choose the safe place

We provide three environment (bedroom, beach, and mountain meadow) and three time periods lighting settings (morning, noon, and afternoon) as shown in Figure1. We did not include the dark light of the night because positive memories are more likely to be associated with bright and warm colors [2]. Here is an excerpt of script in this step: “Start by thinking about a safe place. This might be somewhere you have visited, such as a beautiful beach, forest or mountain meadow. It could also be somewhere that feels safe and comfortable for you, such as a cozy room in your house, or a place in nature where you walk your dog...”.

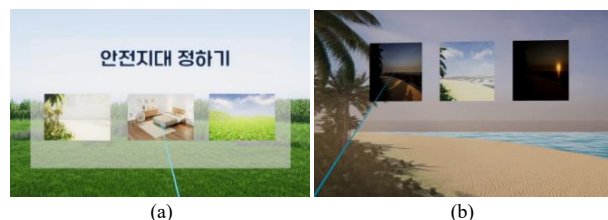


Figure 1: (a) User can choose one of the environments as their safe place; (b) Morning lighting setting in beach environment.

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Figure 2: (a) Comfort item options in bedroom: Scented candles, music player, mug cup, cushion, plants, frames, floor lamps. (b) An example of safe place created in mountain meadow environment; (c) User can take the photo and name the safe place to keep it for future time.

3.2 Create the safe place

In second session, not only the environment representing the safe place, we provide several options of comfort items of different sensory which user can choose and add into the safe place to guide the potential positive memories and emotions, as shown in figure 2(a). Below is an excerpt from the script: *“Explore your safe place, using all of your senses – what can you see, hear, feel, smell, taste and touch? If it’s a beach you could visualize the beautiful turquoise sea, golden sands and blue skies, hear the gulls and breeze rustling palm fronds, feel the sand between your toes...”*.

3.3 Observe the safe place

This session is the practice phase that allow user to take their time to look around and observe the entire place created by themselves. The comfort items they added in previous session can serve as attention anchors for the safe place, as shown in figure 2(b). Below is an excerpt from the script in this step: *“Think back to a recent episode in which you felt uncomfortable or stressed. As you recall that, let’s focus once again on the safe place in front of you. Focus on the appearance, sound, smell, and body sensations of this place where you can rest comfortably and safely...”*.

3.4 Keep the safe place

Since VR offers the advantages of visualization and interactivity, we additionally provide a virtual camera effect, enabling users to capture their safe spaces as images, rather than keeping them in their minds as in traditional therapy.

As the final session, user can take a picture from any angle they want and name their safe place to keep it, as shown in figure 2(c). Below is an excerpt from script which is original script for this step: *“For the last step, take a picture for your safe place. You will be able to take it out whenever you want. Then, name the safe place. Give it a short but simple name that you can easily think of. Now, when you experience unpleasant or uncomfortable feelings, you can visit anytime and relax.”*

4 CONCLUSION AND FUTURE WORK

We introduced Safe Place VR, a trauma stabilization technique based on guided imagery (GI). It is designed to help users create a mental image of a safe space through VR, promoting a sense of safety and calm. The technique consists of four practice steps: selecting, creating, observing, and keeping the safe place.

For the future work, we intend to evaluate the effectiveness of the *Safe Place VR* through user experiments. Additionally, we plan to add multisensory components for comfort item, to enhance the sensory experience of GI. We believe this approach shows the potential to extend into broader mental health and medical research, further address the limitations of traditional therapies.

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